Practical-7

Problem Definition:

Design and implement a 4-bit synchronous counter that follows the non-linear custom sequence:

$2 \rightarrow 5 \rightarrow 7 \rightarrow 11 \rightarrow 3 \rightarrow 2 \rightarrow \ldots$ (repeating)

Requirements:

1. Draw the state diagram and state transition table.

2. Use any one type of flip-flop (JK, T, or D) for the design.

3. Derive flip-flop input equations using the respective excitation table.

4. Implement the complete synchronous circuit and verify the repeating sequence using simulation software or hardware.

Tools/Technology To Be Used:

- Flip-flops (D/JK/T)
- Logic gates
- Simulation software: Logisim / Multisim / Proteus / TinkerCad / Digital
- Breadboard & ICs (optional for hardware testing)

Aim:

To design and implement a 4-bit synchronous counter that follows a non-linear custom sequence (2 → 5 → 7 → 11 → 3 → 2 → …) using flip-flops and logic gates. The objective includes deriving the required flip-flop input equations using excitation tables and verifying the correct sequence through simulation or hardware implementation.

T flip flop:

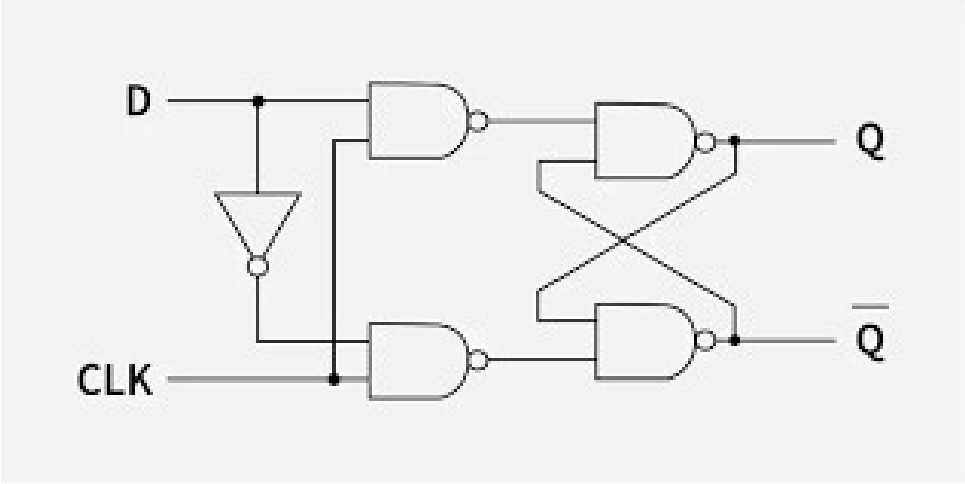
Truth table:

Q(n)	D(n)	Q(n+1) (Next)
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table:

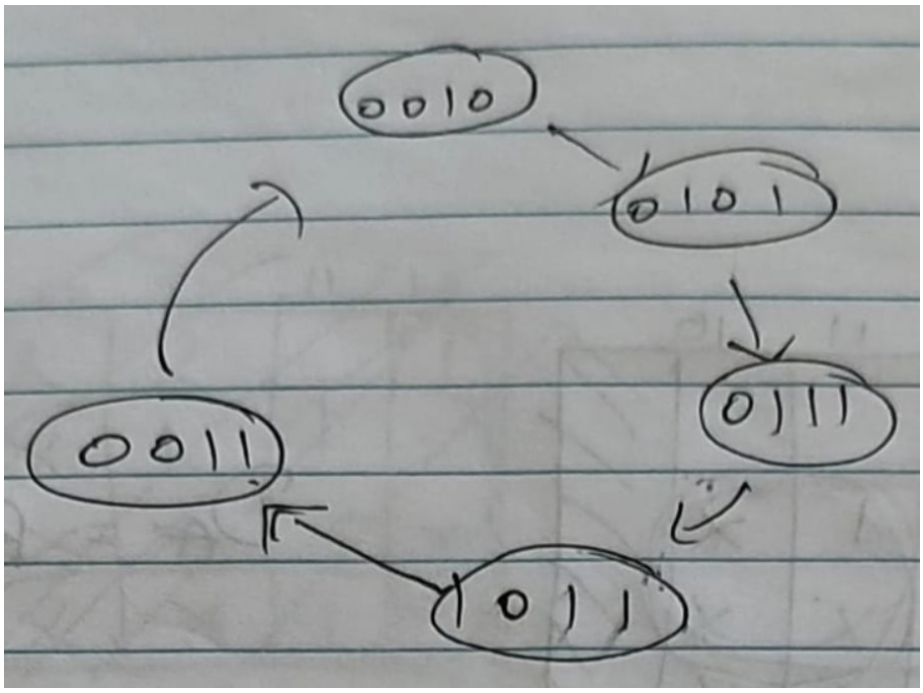
Q Output		Input
$Q_{(n)}$	$Q_{(n+1)}$	D_n
0	0	0
0	1	1
1	0	0
1	1	1

Circuit Diagram:



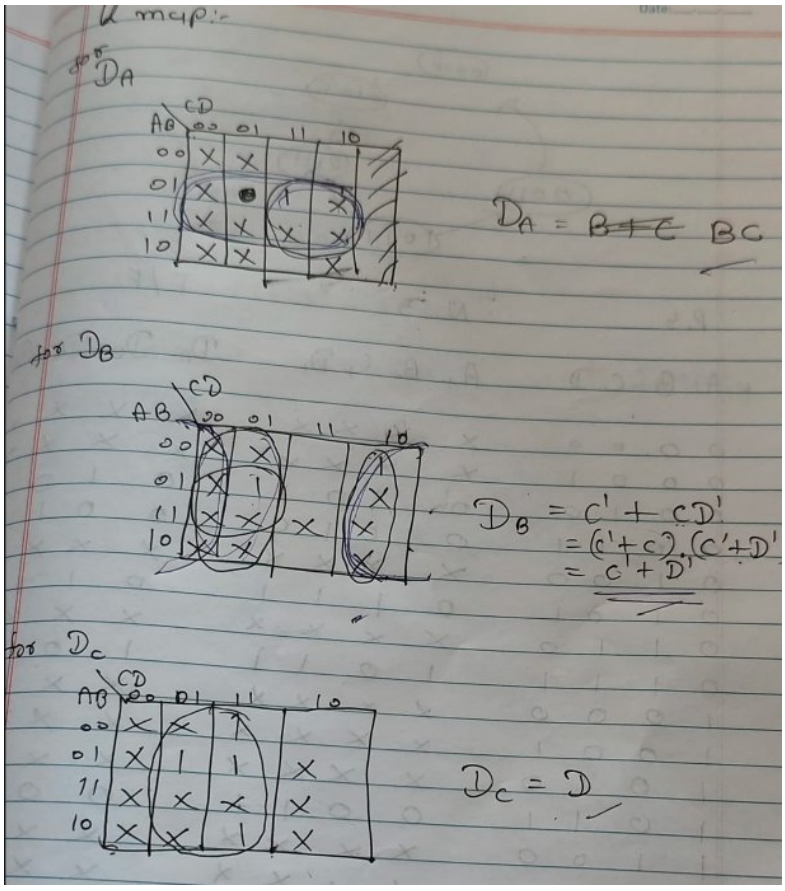
4-bit binary Random counter:

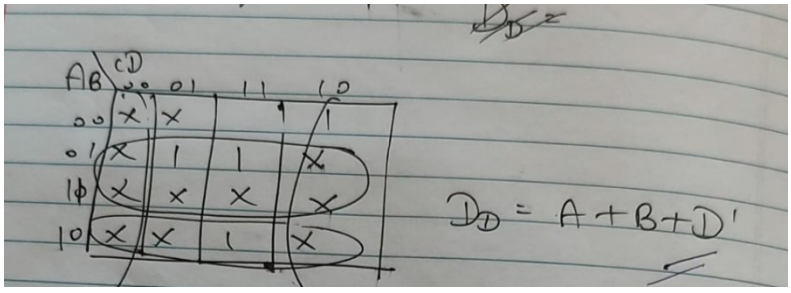
State diagram & State Table:



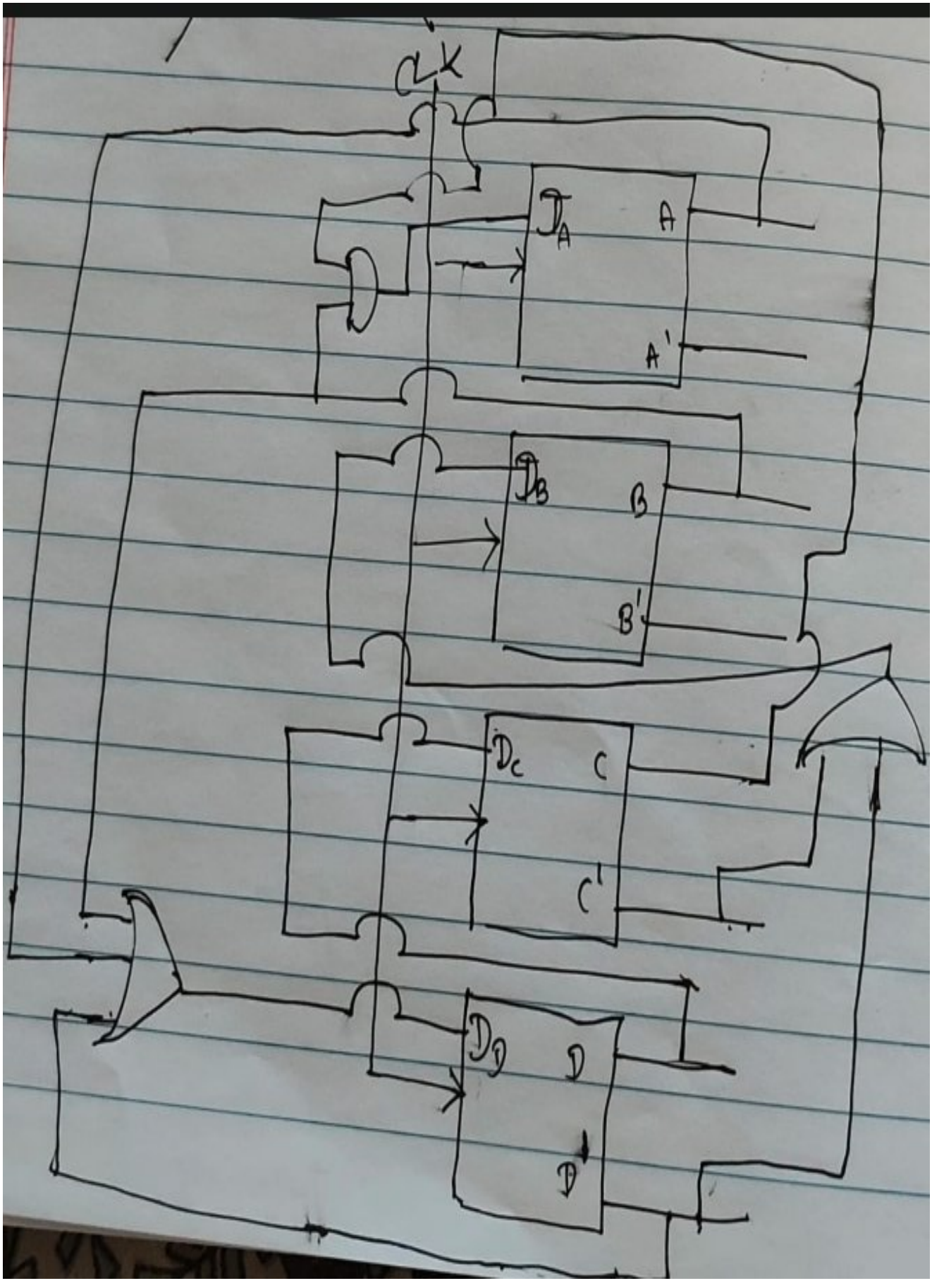
P.S				N.S				F/F			
A	B	C	D	A_+	B_+	C_+	D_+	D_A	D_B	D_C	D_D
0	0	0	0	x	x	x	x	x	x	x	x
0	0	0	1	x	x	x	x	x	x	x	x
0	0	1	0	0	1	0	1	0	1	0	1
0	0	1	1	0	0	1	0	0	0	1	0
0	1	0	0	x	x	x	x	x	x	x	x
0	1	0	1	0	1	1	1	0	1	1	1
0	1	1	0	x	x	x	x	x	x	x	x
0	1	1	1	1	0	1	1	1	0	1	1
1	0	0	0	x	x	x	x	x	x	x	x
1	0	0	1	x	x	x	x	x	x	x	x
1	0	1	0	x	x	x	x	x	x	x	x
1	0	1	1	0	0	1	1	0	0	1	1
1	1	0	0	x	x	x	x	x	x	x	x
1	1	0	1	x	x	x	x	x	x	x	x
1	1	1	0	x	x	x	x	x	x	x	x
1	1	1	1	x	x	x	x	x	x	x	x

K-Map & circuit diagram:

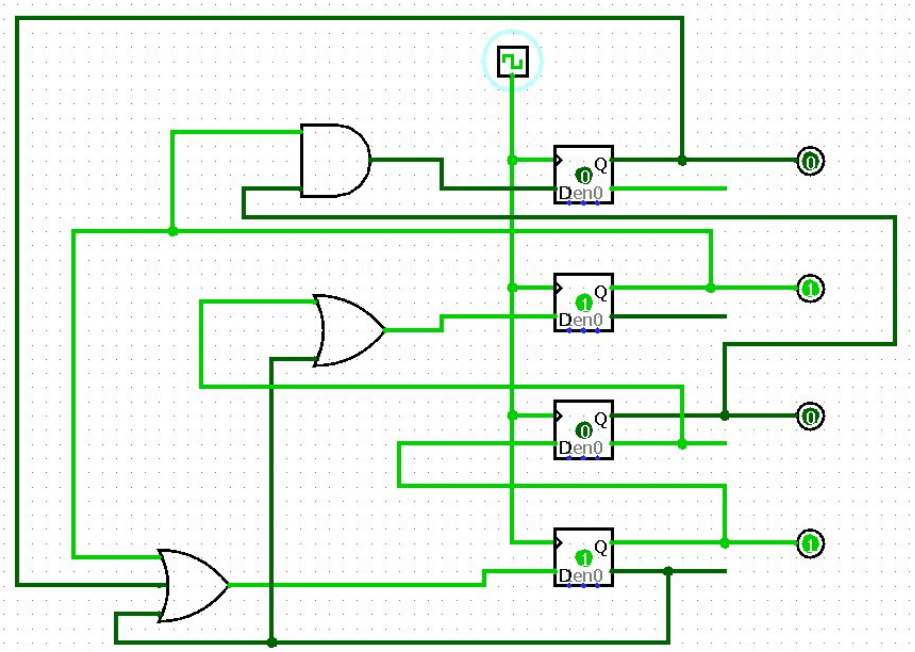


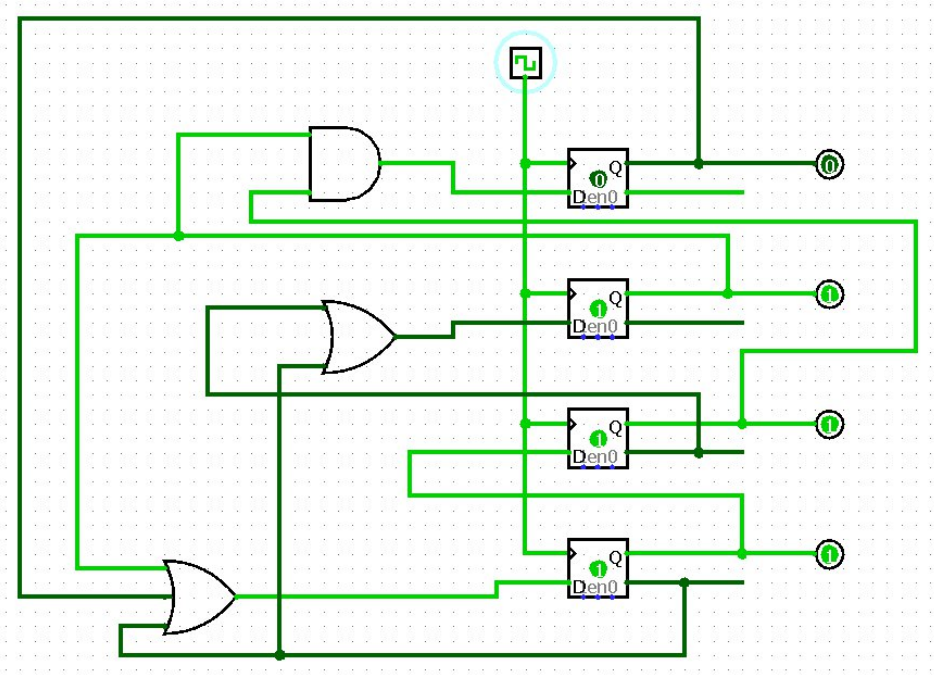


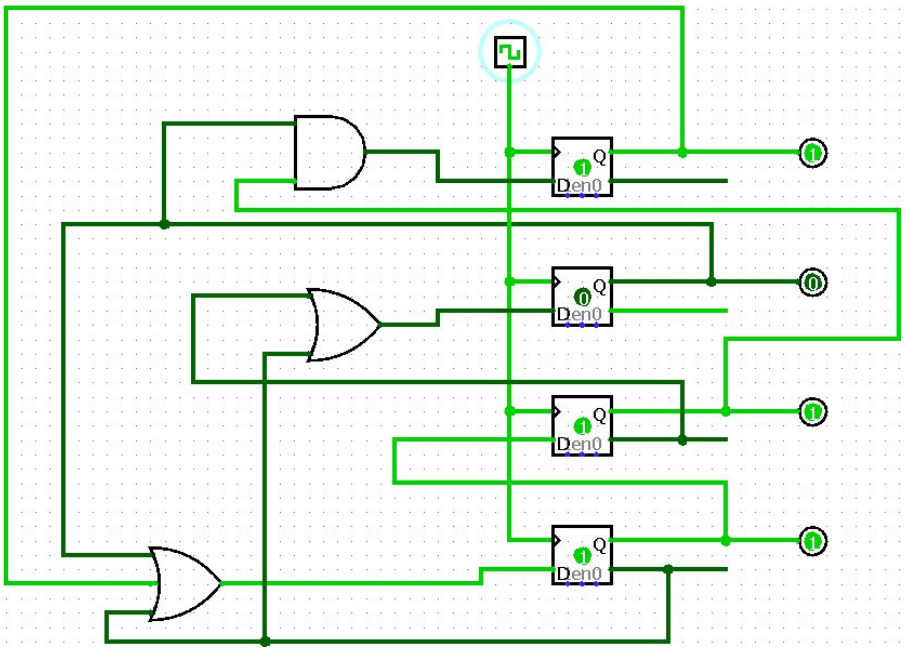
Circuit :

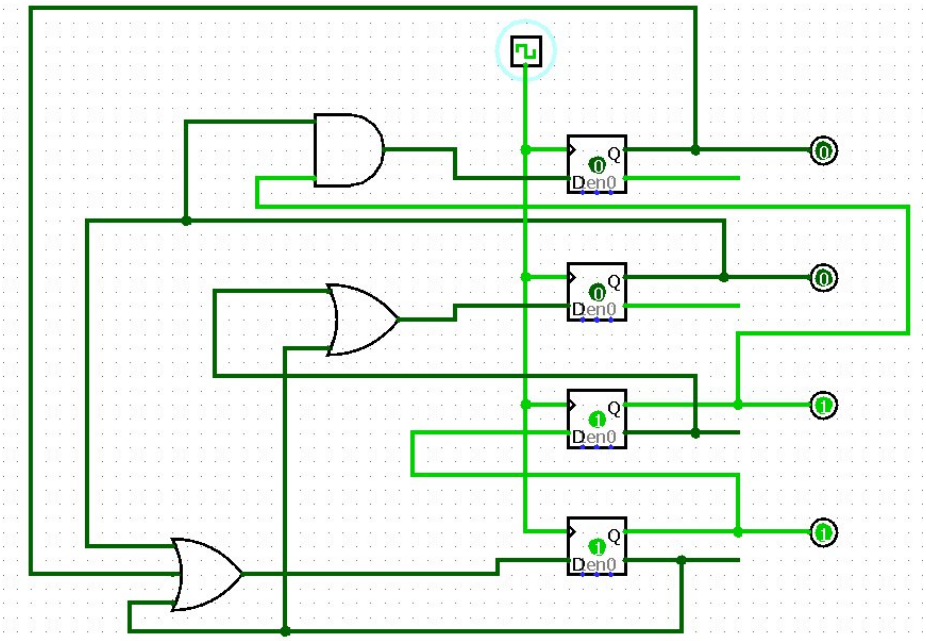


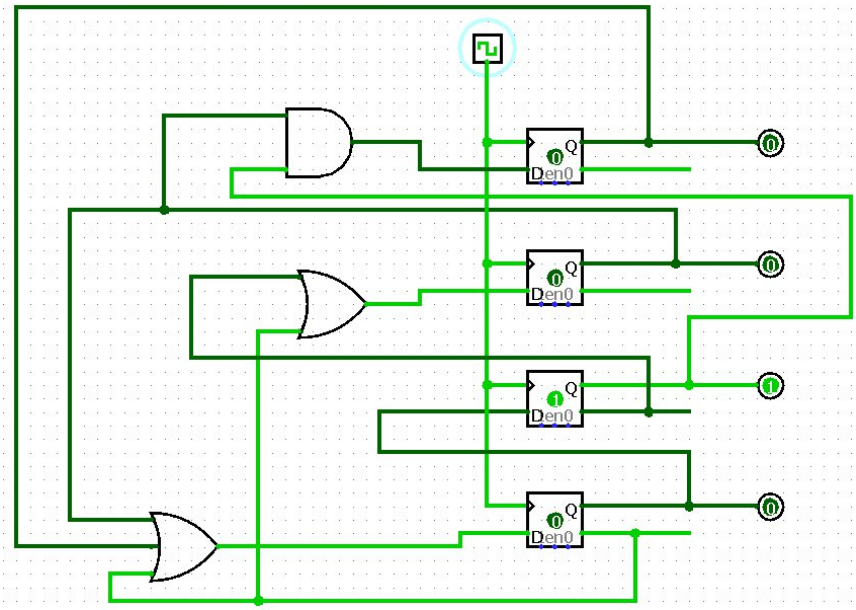
Screen shot:






Q
Den0
Q
Den0
Q
Den0
Q
Den0


Q
Den0
Q
Den0
Q
Den0
Q
Den0



Conclusion : *This practical successfully demonstrates the design of a 4-bit synchronous counter tailored for a non-linear custom sequence. By utilizing state transition tables and flip-flop excitation logic, the circuit proves that synchronous systems can be programmed to skip traditional binary increments in favor of specific numerical patterns ($2, 5, 7, 11, 3$). The implementation confirms that by tying all flip-flop clock inputs to a common source, the system eliminates ripple delays and ensures all state changes occur simultaneously. Ultimately, the project validates the versatility of sequential logic design, showing how K-map minimization can transform a complex counting requirement into an optimized, hardware-efficient circuit.*